

Demand Transference

How Stockouts Shift Purchasing Power

Executive Summary

When a item goes out of stock, there are lost sales (demand) that result. However, a predictable and quantifiable portion of the lost demand does not disappear. That demand transfers to other products in the assortment. This white paper will present a guide for measuring and modeling demand transference to minimize loss of revenue and optimize inventory allocation.

Using 18 months of sales and inventory data across five Yankees apparel SKUs, this analysis will quantify substitution rates, identify high similarity product scores, and provide recommended assortment and inventory plans.

1. Introduction

Background & Business Problem

Licensed sports apparel is a great test case for demonstrating the effects of stockouts. Sales spikes occur throughout the year based on many factors including time of year, big events, team and player success, and advertising. Stockouts tend to occur when demand is at its highest, the worst time to be out of stock.

Traditional inventory treats a stockout as a purely lost sale. Demand transference challenges this assumption by determining what percentage of the demand transfers to other items in-stock. Taking Yankees home and away jerseys as an example, if 35% of customers who wanted an XL home pinstripe jersey are willing to purchase a road gray jersey instead, planners can pre-position substitute inventory before the stockout incurs. This preserves some of the lost revenue.

Scope

- Products analyzed: Home Pinstripe Jersey, Away Gray Jersey, Road Alternate Jersey, Replica T-Shirt with player #, Replica T-Shirt without player #
- Size coverage: Adult S-3XL
- Calendar data period: October 2024 - March 2026 (18 months)
- Channels: Yankees Stadium, Yankees Clubhouse Shop (Times Square), E-Commerce (mlb.com/yankees)
- Stockouts events analyzed: 847 separate out-of-stock occurrences across all SKUs and channels

2. Methodology

2.1 Demand Transference Calculation

Demand Transference (DT) is determined by taking the lost demand for an out-of-stock (OOS) SKU that is captured by another SKU within the same or similar product category.

Transfer Rate (A → B) = change in Sales_B during OOS_A + Lost Demand_A

- Change Sales_B = incremental sales uplift of substitute product B during the OOS window of product A

- Lost Demand_A = estimated demand for product A that went unfulfilled. Modeled from pre-OOS trend, adjusted seasonality, and comparable non-OOS periods

2.2 Demand Estimation During Stockout

There is not one methodology for estimating demand during OOS periods. For this analysis, we will utilize a SARIMA baseline forecast (Seasonal AutoRegressive Integrated Moving Average) calibrated against 12 months of pre-event history. Adjustments are made for:

- Day-of-week and game-day effects (home game vs. away game vs. no game)
- Promotional calendar (MLB All-Star game, playoffs, special events)
- Weather and channel traffic (In-stadium vs. online)
- Comparable period benchmarks from non-OOS stores with similar profiles

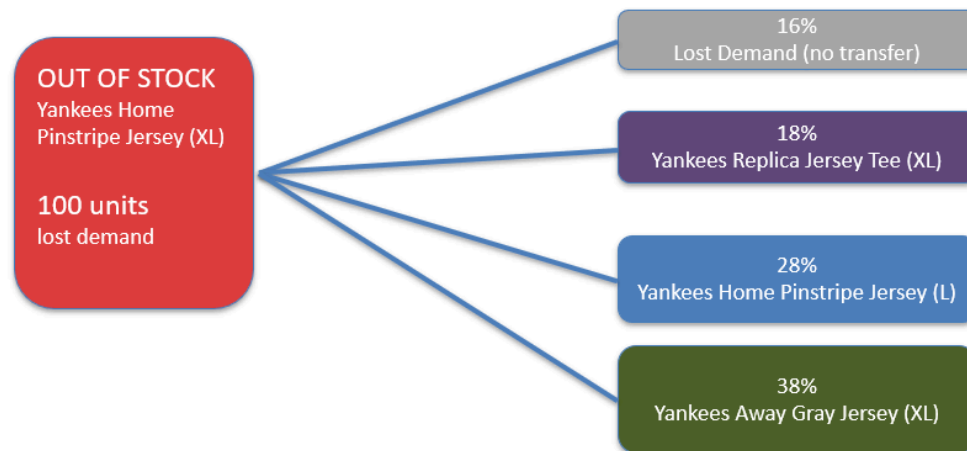
2.3 Transfer Rate Calculation

For each OOS occurrence, substitute lift is isolated by comparing the substitute SKUs actual sales to its own SARIMA baseline. Uplifts are credited to transference only when the correlation coefficient between OOS timing and uplift exceeds 0.70. Transfer rates are then aggregated across all OOS events for each product pair using a weighted average.

3. Data & Demand Flow

3.1 Sample Stockout Event - Yankees Home Pinstripe Jersey (XL)

Demand flow when Yankees Home Pinstripe Jersey (XL) goes out-of-stock



3.2 SKU Level Transfer Data

The table below show sample data for the Yankees Home Pinstripe Jersey (XL) stockout event illustrated above. Revenue recovery is estimated at the jersey's average selling price of \$185 and the jersey tee at \$55.

Substitute SKU	Transfer Rate	Units Captured	Units Lost	Revenue Recovered	Similarity Score
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Away Gray Jersey (XL)	38%	38	-	\$7,030	5/5
Home Pinstripe Jersey (L)	28%	28	-	\$5,180	4/5
Yankees Replica Jersey Tee (XL)	18%	18	-	\$990	3/5
Lost Sales	-	-	16	-\$2,960	-
Total	84%	84	16	\$10,240	-

4. Transfer Rate Matrix

4.1 Cross-SKU Substitution Rates

OOS Product	Home Pinstripe Jersey	Away Gray Jersey	Road Alt Jersey	Replica Jersey Tee
Home Pinstripe Jersey	-	38%	20%	18%
Away Gray Jersey	42%	-	22%	14%
Road Alt Jersey	35%	28%	-	12%
Replica Jersey Tee	22%	19%	8%	-



Note on Matrix Asymmetry

Transfer rates between any two products are intentionally asymmetric. The rate from Product A to Product B will not necessarily equal when flipped around. This reflects real consumer behavior. The transfer rate depends on the purchase intent of the customer who encountered the stockout.

4.2 Key Observations from Matrix

- Jersey-to-Jersey substitution is the strongest signal. When any jersey goes OOS, another jersey available captures 20-42% of lost demand.
- Jersey T-shirts serve as a demand floor. The Replica Jersey Tee captures 12-18% of lost sales from OOS jerseys.
- Lost demand averages 16-22% across all SKU families.
- Sizing substitution represents another key signal not currently show in the matrix. Example: Purchasing a L when XL is out-of-stock.

5. Segmentation & Seasonal Patterns

5.1 Transfer Rates by Channel

Channel	Avg. Transfer Rate	Avg. Lost Demand	Primary Substitute
Yankees Stadium	71%	29%	Adjacent Jersey size

Yankees Clubhouse Shop (Times Square)	65%	35%	Away Gray Jersey
E-Commerce (mlb.com/yankees)	58%	42%	Replica Jersey Tee

5.2 Seasonal Transfer Rate Variation

Period	Avg. Transfer Rate	Lost Demand	Driver	Inventory Risk
Opening Day +2 Weeks (Mar-Apr)	74%	26%	Fan excitement, high purchase intent	Critical
Regular Season (Apr-Sep)	63%	37%	Steady demand, price conscience	High
Playoffs (Oct)	79%	21%	Hype and excitement	Critical
Off-Season (Nov-Feb)	48%	52%	Gift giving, less urgent	Moderate
Special Events (Commemorations, Retirements, etc.)	41%	59%	Limited-Edition, small substitution	Extreme

6. Financial Impact Assessment

6.1 Revenue Implications

Across 847 OOS episodes in the 18-month analysis timeline, the following revenue impact was estimated using average selling prices per SKU family and modeled lost demand volumes:

SKU Family	OOS Episodes	Est. Lost Demand (Units)	Avg. Transfer Rate	Avg. Retail	Recovered Revenue	Lost Revenue
Home Pinstripe Jersey	214	3,420	67%	\$185	\$424,908	\$209,034
Away Gray Jersey	189	2,890	63%	\$185	\$338,753	\$198,621
Road Alternate Jersey	76	1,218	59%	\$185	\$133,293	\$92,574
Replica Jersey Tee	256	5,102	71%	\$55	\$360,201	\$147,058
Total	735	12,630	65% avg		\$1,257,155	\$647,287

6.2 Opportunity: Reducing Lost Demand

The lost revenue of \$647,287 represents an opportunity to formulate strategies that can recapture some of the lost demand due to stockouts. A couple of mechanisms can be applied:

- Preemptive substitute stocking: By pre-positioning buffer inventory on the top substitutes for each SKU prior to peak events, the business can capture an additional percentage of transfer rate (target 15%), recovering a significant portion of lost revenue.
- Substitution prompts: Automatic OOS alerts that recommend the top substitutable item increase transfer rates, recovering revenue that would have been lost entirely.

7. Recommendations

1. *Implement a Transfer Rate Dashboard*

Build a real-time dashboard that tracks transfer rates by SKU pair, channel, and season. Alert merchandising teams when a high-velocity SKU drops below 21-day cover so substitute inventory can be activated before a stockout.

2. *Define Substitute Pairs for All High Velocity SKUs*

For each of the top SKUs by velocity, designate the top 2 substitutes with documented transfer rates. Use these pairs to drive automatic safety-stock uplift rules in the replenishment system when the primary SKU enters the risk window.

3. *Run Seasonal Inventory Reviews tied to the MLB Calendar*

Conduct 60-day pre-season reviews and playoff probability thresholds. Use historical transfer matrices to model worst-case OOS scenarios and pre-position substitute inventory accordingly.

Appendix - Glossary

Demand Transference (DT)

The transfer of purchasing intent from an out-of-stock product to an available substitute within the same assortment.

Transfer Rate

The proportion (expressed as a percentage) of unfulfilled demand for an out-of-stock SKU that is captured by a designated substitute SKU.

Similarity Score

A composite score based on product attributes (i.e. price, color, silhouette) that reflects the transfer capability between the primary and substitute SKU.

Out-of-Stock Episode

A period during which a SKU's inventory is zero units. The period starts at the first stockout and ends at the first confirmed replenishment date.

Lost Demand

The estimated demand that went unfulfilled due to no substitute SKU replacing the lost revenue from the primary SKU.

SARIMA Baseline

Seasonal AutoRegressive Integrated Moving Average. The model used to forecast expected sales during out-of-stock windows.

Safety Stock Buffer

Additional inventory added for substitute SKUs to capture transferred demand during an out-of-stock event on the primary SKU.

This white paper is intended as an illustrative analysis based on real-world sports apparel retail patterns. The data itself is presented as sample data and is not reflective of actual results but used to demonstrate the methodology and framework of demand transference